



AstraQuantum Tech

AstraQuantum Tech

Week 1 Cybersecurity Assignment

Hands-on TryHackMe Labs - Foundations of Operating Systems, the Web, and Cybersecurity Careers

Name	Muhammad Raza
Program	AstraQuantum Tech - Cybersecurity Summer Programme
Platform	TryHackMe
Rooms Completed	5 / 5 (100%)

Table of Contents

Assignment Overview.....	3
Room 1: Getting Started	4
Room 2: Careers in Cyber	5
Room 3: HTTP in Detail.....	6
Room 4: Linux Fundamentals Part 1.....	9
Room 5: Windows Fundamentals Part 1.....	10
Conclusion	12

Assignment Overview

This report documents the completion of the Week 1 Cybersecurity Assignment issued by AstraQuantum Tech. The objective of this week's task was to build a strong foundation in cybersecurity by completing five hands-on labs on the TryHackMe platform, covering the fundamentals of operating systems, the web, and cybersecurity career paths.

Task Instructions

- ✦ Solve the Rooms: Complete all 5 assigned rooms listed below step-by-step.
- ✦ Create Documentation: For each room, a short summary/documentation is provided covering what was learned, key concepts/commands/takeaways, and screenshots of progress or completed tasks.
- ✦ Profile Tracking: The lead instructor/team profile on TryHackMe (muhammadoxahmed) has been followed so activity and performance can be monitored for upcoming CTF teams.

Assigned Rooms

- ✦ Getting Started - Learn how to navigate the TryHackMe platform and get started with your first steps.
- ✦ Careers in Cyber - Explore different career domains in cybersecurity to find where your passion lies.
- ✦ HTTP in Detail - Understand how the web works under the hood by diving into HTTP requests, responses, and headers.
- ✦ Linux Fundamentals Part 1 - Learn essential Linux terminal commands and file navigation, which are core skills for any security professional.
- ✦ Windows Fundamentals Part 1 - Understand the basics of the Windows operating system and its core security features.

Room 1: Getting Started

What I Learned

Learned how to navigate the TryHackMe platform and completed my first steps.

Key Concepts, Commands, or Takeaways

- ✦ Inspecting web page source code to uncover hidden information.
- ✦ Identifying HTML comments, which begin with the <!-- character.
- ✦ Discovering hidden directories or pages (like /test-admin) left behind by developers in the source code.
- ✦ Testing common default credentials, such as admin:admin, to gain unauthorized access.
- ✦ Navigating an admin portal to extract information, such as viewing the list of signed-up users to determine the total user count.

Proof of Completion

The screenshot shows the TryHackMe interface for the 'Getting Started' room. The top navigation bar includes links for Dashboard, Learn, Practice, Compete, and HH 2026, along with a search icon, a notification bell, a 'Go Premium' button, and user statistics (34 diamonds, 1 flame). The main content area features a 'Getting Started' title with a subtitle 'Get started with TryHackMe by hacking a fake social media website!'. Below this, a progress bar indicates 'Room completed: 100%'. A list of tasks is shown, all marked as completed with green checkmarks: Task 1 Introduction, Task 2 Default Credentials, and Task 3 Conclusion. The bottom of the page shows a list of tasks with their completion status.

TryHackMe

Dashboard Learn Practice Compete HH 2026

Go Premium 34 1

← Back to all walkthroughs

Getting Started

Get started with TryHackMe by hacking a fake social media website!

45 min 26,717

Share your achievement Start AttackBox Save Room 687 Recommend Connectivity Options

Room completed: 100%

Task 1 Introduction

Task 2 Default Credentials

Task 3 Conclusion

Room 2: Careers in Cyber

What I Learned

Explored a comprehensive range of career domains within the cybersecurity field to understand the landscape of industry roles and specializations.

Key Concepts, Commands, or Takeaways

- ✦ Understanding the daily responsibilities of a Security Analyst and Incident Responder, which align closely with active Blue Team operations, threat monitoring, and phishing analysis.
- ✦ Exploring the Security Engineer role, mapping directly to the design, configuration, and troubleshooting of secure, enterprise-level network topologies.
- ✦ Identifying the investigative scope of a Digital Forensics Examiner and Malware Analyst when analyzing breaches and reverse-engineering malicious code.
- ✦ Differentiating between a Penetration Tester and a Red Teamer, tying into the offensive security strategies and methodologies utilized in CTF competitions.

Proof of Completion

The screenshot shows the TryHackMe interface for the 'Careers in Cyber' room. The top navigation bar includes links for Dashboard, Learn, Practice, Compete, and HH 2026, along with a search icon, a bell icon, a 'Go Premium' button, and user statistics (34 diamonds, 1 flame). The room title 'Careers in Cyber' is prominently displayed, with a subtitle 'Learn about the different careers in cyber security.' Below the title, it indicates a duration of 30 minutes and a popularity of 1,283,405 views. A row of action buttons includes 'Share your achievement', 'Save Room', '15534 Recommend', 'Connectivity', and 'Options'. A green progress bar indicates 'Room completed: 100%'. The task list on the left shows three tasks, all marked as completed with green checkmarks: 'Task 1 Introduction', 'Task 2 Security Analyst', and 'Task 3 Security Engineer'.

Room 3: HTTP in Detail

What I Learned

Explored how the web works under the hood by examining the structure of HTTP requests and responses, including how clients and servers communicate data securely.

Key Concepts, Commands, or Takeaways

- ✦ **HTTP vs. HTTPS:** Understood the basics of HyperText Transfer Protocol and the importance of the “secure” (S) certificate distinction.
- ✦ **HTTP Methods:** Identified the core actions used to interact with web servers: GET (viewing/retrieving data), POST (creating records), PUT (updating records), and DELETE (removing records).
- ✦ **HTTP Status Codes:** Learned how to interpret server responses, such as 201 (Created), 401 (Unauthorized), 404 (Not Found), and 503 (Service/Database Unavailable).
- ✦ **HTTP Headers:** Explored key headers that provide context to requests and responses, including Content-Length (data size), User-Agent (browser identification), Content-Type (data format), Host (requested website), and Set-Cookie (for storing session data).
- ✦ **Practical Application:** Successfully crafted custom requests by modifying URI parameters and body parameters (like username and password) to authenticate and manipulate endpoints.

Proof of Completion

The screenshot shows the 'HTTP in Detail' room completion screen on the TryHackMe platform. The top navigation bar includes the TryHackMe logo, a 'Go Premium' button, and user statistics (34 diamonds, 1 flame). The main content area features a 'Back to all walkthroughs' link, a laptop icon, the title 'HTTP in Detail', and a description: 'Learn about how you request content from a web server using the HTTP protocol'. Below this, it shows a progress bar, a timer for 30 minutes, and a user count of 669,536. A row of buttons includes 'Share your achievement', 'Badge', 'Save Room', '8974 Recommend', 'Connectivity', and 'Options'. At the bottom, a green bar indicates 'Room completed: 100%'.

Room 4: Linux Fundamentals Part 1

What I Learned

Learned essential Linux terminal commands and file navigation techniques required to operate effectively via the command line.

Key Concepts, Commands, or Takeaways

- ✦ **echo / whoami:** Outputting text to the terminal using the echo command and identifying the current logged-in user with the whoami command.
- ✦ **cd / pwd:** Navigating the file system using cd (change directory) and determining the absolute path of the current directory using pwd (print working directory).
- ✦ **grep:** Searching for specific text strings or patterns within files, such as locating flags in log files, using the grep command (e.g. `grep -R "PRETTY_NAME" /etc/`).
- ✦ **Background execution:** Running commands in the background by appending the & operator.
- ✦ **Redirection operators:** Managing file outputs with redirection operators — using > to overwrite file contents and >> to append new data to an existing file without deleting its previous contents.

Proof of Completion

The screenshot shows the 'Linux Fundamentals Part 1' room completion page on TryHackMe. The top navigation bar includes 'Dashboard', 'Learn', 'Practice', 'Compete', and 'HH 2026'. The main header features the TryHackMe logo and a 'Go Premium' button. Below the header, a 'Back to all walkthroughs' link is visible. The room title 'Linux Fundamentals Part 1' is prominently displayed, accompanied by a penguin icon. A description reads: 'Embark on the journey of learning the fundamentals of Linux. Learn to run some of the first essential commands on an interactive terminal.' Below this, icons represent a book, a terminal, a clock (20 min), a group of people (1,207,263), and a globe. A row of action buttons includes 'Share your achievement', 'Badge', 'Save Room', '18463 Recommend', 'Connectivity', and 'Options'. At the bottom, a green bar indicates 'Room completed: 100%'.

Room 5: Windows Fundamentals Part 1

What I Learned

Explored the basics of the Windows operating system, focusing on its core architecture, administrative tools, and foundational security features.

Key Concepts, Commands, or Takeaways

- ✦ **Windows GUI & File System:** Navigated the desktop environment and examined the underlying directory structure.
- ✦ **System Architecture:** Explored the Windows\System32 directory, which houses the critical executables and libraries required for the operating system to function.
- ✦ **Access Management:** Reviewed User Accounts, Profiles, and Permissions to understand how Windows restricts or grants access to files and system resources.
- ✦ **Administrative Controls & Monitoring:** Examined User Access Control (UAC) mechanisms for preventing unauthorized system changes and utilized the Task Manager for process and performance monitoring. Reviewing these specific administrative controls reinforces the foundational troubleshooting principles established in the Google IT Support Professional Certificate.
- ✦ **System Configuration:** Managed OS settings through the Settings app and the legacy Control Panel.

Proof of Completion

The screenshot shows the 'Windows Fundamentals 1' room completion page on TryHackMe. The top navigation bar includes 'TryHackMe', 'Dashboard', 'Learn', 'Practice', 'Compete', and 'HH 2026'. A search bar, a 'Go Premium' button, and user statistics (34 diamonds, 1 flame) are on the right. The main content area features a large Windows logo and the title 'Windows Fundamentals 1'. Below the title, it states: 'In part 1 of the Windows Fundamentals module, we'll start our journey learning about the Windows desktop, the NTFS file system, UAC, the Control Panel, and more..'. It also shows a timer for 30 minutes and a user count of 589,627. A row of buttons includes 'Share your achievement', 'Start AttackBox', 'Save Room', '7501 Recommend', 'Connectivity', and 'Options'. A green bar indicates 'Room completed: 100%'. Below this, a list of tasks is shown: 'Task 1 Windows Editions', 'Task 2 The Desktop (GUI)', and 'Task 3 Introduction to Windows', all marked as completed with green checkmarks.

Conclusion

Over the course of Week 1, all five assigned TryHackMe rooms were completed successfully, establishing a solid foundation across three critical pillars of cybersecurity: platform fundamentals, career awareness, and core technical skills.

The Getting Started and Careers in Cyber rooms provided the orientation needed to navigate the TryHackMe platform confidently while mapping out the wide range of career paths available in the industry - from Security Analysts and Engineers to Penetration Testers and Digital Forensics Examiners. The HTTP in Detail room built a practical understanding of how web communication works, including request methods, status codes, and headers, which forms the backbone of most web application security work.

The Linux Fundamentals Part 1 and Windows Fundamentals Part 1 rooms delivered hands-on exposure to both major operating system environments used across the security field - from command-line navigation, file redirection, and text searching in Linux, to user access control, system architecture, and administrative tooling in Windows.

Together, these rooms have built the essential groundwork required for the more advanced, offensive- and defensive-security-focused labs expected in the weeks ahead. The lead instructor's TryHackMe profile has been followed as instructed for activity tracking ahead of upcoming CTF team formation.